The Hendrickson Hybrid Design: Response Components, the Gompertz Model, and the Analysis Model

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Purpose. This paper explains how the trial design of Hendrickson et al. [1] works, what its simulation assumes, and how its analysis model relates to that simulation. The design is the open-label, blinded discontinuation and crossover design that the paper calls the 'N-of-1 design' and this project calls Hybrid. The paper brings together material spread across `docs/24`, `docs/25`, `docs/31`, `docs/36`, `docs/37`, `docs/38`, `docs/39` and manuscript 07 (the Gompertz evaluation). It corrects several errors in those documents (Section 11) and adds new Monte Carlo results on the form of the time adjustment (Section 9). Eight figures develop the intuition.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here and are not part of the canonical set.

Notation index

Trial, indices and time.

Table 1: Notation index for trial, index and time symbols

Symbol	Meaning	Status
i, t	participant and visit indices	canonical
N	total participants across paths; 70 throughout	canonical
P	number of randomization paths; 4 for Hybrid	canonical
p	path index	local
w_t	calendar week of visit t (4, 8, 9, 10, 11, 12, 16, 20)	local
t_{od}	time on drug in the current run (code `tod`)	canonical
t_{sd}	time since discontinuation (code `tsd`)	canonical
t_{pb}	time under positive expectancy (code `tpb`)	canonical
$t_{1/2}$	carryover half-life, weeks	canonical
e_t	design expectancy at visit t: 1 open label, 0.5 blinded, 0 at baseline (code `e`, regressor `De`)	local

Outcome and components.

Table 2: Notation index for outcome and response component symbols

Symbol	Meaning	Status
Y_{it}	symptom score (CAPS-IV total; code column `Sx`)	canonical
BL_i	baseline symptom score	canonical
BR_{it}	biological (drug) response; Hendrickson's BR	canonical
PB_{it}	placebo-belief (expectancy) response; Hendrickson's ER	canonical
TV_{it}	time-variant natural-course response; Hendrickson's TR	canonical
u_i, ε_{it}	participant random intercept and residual	canonical
$G(t; m, d, r)$	modified Gompertz curve, Section 4	local
m, d, r	ceiling, displacement and rate of a component's curve (code `max`, `disp`, `rate`)	local
t^*	inflection time of the Gompertz curve, $\ln(d)/r$	local
μ_{BL}	population mean of the baseline score (83.07)	local

Symbol	Meaning	Status
$TV(w_t)$, $PB(t_{pb})$, $BR_p(t)$	mean trajectories; BR_p in path p	local

Biomarker and moderation.

Table 3: Notation index for biomarker and moderation symbols

Symbol	Meaning	Status
B_i, b_i	biomarker (standing systolic blood pressure) and its standardized value	canonical
σ_{BR}, σ_{bm}	standard deviations of BR (8) and of the biomarker (15.36)	canonical
c_{bm}	covariance-moderation strength: correlation of B with BR where coupled	canonical
β_{bm}	mean-moderation multiplier	canonical
g_t	coupling at visit t: fraction of c_{bm} applied there	local
$\bar{g}_{\text{on}}$, $\bar{g}_{\text{off}}$	mean coupling over a path's on-drug and off-drug visits	local
$\overline{\text{gap}}_{\text{on}}$, $\overline{\text{gap}}_{\text{off}}$	biomarker band width (± 1 SD) averaged over on-drug and off-drug visits	local
ρ	within-factor correlation, 0.8 at every lag (compound symmetry, CS); canonical symbol, used here for the CS rather than the AR(1) correlation	canonical
c_1, $c_\times$	cross-factor correlation at the same visit (0.2) and at different visits (0.1)	local

Analysis model.

Table 4: Notation index for analysis model symbols

Symbol	Meaning	Status
D_{it}	binary drug state (code `Db`)	canonical
$D_{bc,it}$	exposure-decayed drug indicator (code `Dbc`)	canonical
β_0, β_B	intercept and biomarker main effect	local (β_B)
β_t, β_e	coefficients of linear time and of expectancy	local
τ_t	visit effect when time enters as a factor	local
β_D	drug main effect	canonical
$\beta_{bm:D}$	biomarker-by-drug interaction, the estimand	canonical
α	test size (0.05)	canonical
R^2	squared multiple correlation of `De` on `Db` and `t`	local

Sign convention. The three components are reductions in symptom severity, so they enter Y with a negative sign. Drug effects and interaction coefficients are therefore negative, and the moderation parameters c_{bm} and β_{bm} are positive.

Glossary

- **Hybrid.** The Hendrickson 'N-of-1' design: 8 weeks open label, a 4-week blinded discontinuation block, then a blinded two-period crossover (Section 2).
- **Path.** One sequence of drug states through the trial. Hybrid has $P = 4$, from two randomizations.
- **Blinded discontinuation (BD).** A block in which participants know they may be switched to placebo but not when. In Hybrid everyone stops during it; only the timing is randomized.
- **Coupling.** The rule that sets g_t, that is, at which visits the biomarker is correlated with BR and how strongly. *Step* (published): $g_t = 1$ wherever the mean of BR is nonzero. *Graded*: 1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug.
- **Covariance moderation.** The interaction is carried by the correlation between B and BR (the published architecture). **Mean moderation.** The interaction is a shift of BR by $\beta_{bm}\sigma_{BR}b_i$ at on-drug visits.
- **E9.** The paired-difference statistic of `docs/37`: each participant's on-drug mean minus off-drug mean, regressed on the biomarker. Its slope has the same expectation as $\beta_{bm:D}$.
- **Tabula rasa (TaRa).** Hendrickson's assumption that TV and PB share the same Gompertz parameters and variance.
- **Problem 1 and problem 2.** The two known defects of the published simulation (`docs/36`). Problem 1: correlation matrices that are not positive definite, which the code repairs without saying so. Problem 2: the collapse of power under negligible carryover.

1. Summary

1. **The design answers a question about individuals, not arms.** The estimand is whether baseline standing blood pressure predicts the size of a participant's response to prazosin. Every participant is on drug and off drug at some point, so each participant contributes a within-person contrast (Section 2).
2. **Three response components, each on its own clock.** Drug response runs on time on drug, expectancy response on time under expectancy scaled by the blinding level, and natural course on calendar time. In the Hybrid design TV and PB are identical in every path: all paths share the visit schedule and the blinding pattern. Only BR differs between paths (Section 3, Figure 3).

3. **The Gompertz curve is a slow-start, saturating rise.** At the published parameters the drug response reaches 3% of its ceiling after one week, 39% after four and 84% after eight. Re-exposure starts the curve from zero, so a crossover period of four weeks on drug recovers only 39% of the ceiling (Section 4, Figure 2).
4. **The interaction is how much wider the biomarker band is on drug than off drug.** Drawn on Hendrickson's trajectory template, the expected trajectories of high- and low-biomarker participants separate on drug and close off drug. The estimand is the difference in that separation, divided by $2\sigma_{bm}$. Under the published step rule, any carryover makes the band equally wide off drug, so the estimand is exactly zero. That is problem 2 (Section 5, Figures 4 and 5).
5. **The analysis model is deliberately simpler than the simulation.** The published model is `Sx ~ bm + Db + t + bm*Db + (1|ptID)`. Its linear `t` stands in for TV and PB. The expectancy term `De` exists in the code but was switched off, because it was collinear with `Db`: their correlation in the Hybrid long data is 0.59 (Section 8).
6. **The time adjustment changes the drug effect, not the interaction.** Linear `t`, a spline and visit-as-factor give the same mean interaction estimate, within 0.0007. For the drug main effect they differ by about 2 points in Hybrid, because linear time cannot absorb the expectancy drop at blinding and attributes the extra open-label improvement to the drug. A population-mean calculation reproduces the simulated estimates (Section 9, Figure 7).
7. **Flexible time adjustment is not a free gain in power.** Spline and visit-as-factor raise interaction power by 0.02 to 0.07 in Hybrid, but they also raise the Type I error. Under one-week carryover it reaches 0.073 and 0.080, against 0.058 for linear `t`. In the crossover all three forms hold their size (Section 9, Figure 8).

2. The trial and its question

Hendrickson et al. designed the simulation to plan a trial of prazosin for post-traumatic stress disorder (PTSD), registered as NCT03539614. The hypothesis was that standing systolic blood pressure at baseline predicts the reduction in PTSD symptoms that prazosin produces. That relationship had been seen in a post hoc analysis of an earlier parallel-group trial [1, 2]. The estimand is therefore not a treatment effect but a treatment-by-biomarker interaction: whether participants with higher blood pressure respond more to the drug.

A parallel-group trial answers this question poorly, because a participant on placebo carries no information about their own drug response. A design in which every participant is observed both on and off drug supplies a within-person contrast for each participant, and the interaction becomes a question about how those contrasts vary with the biomarker. Hendrickson et al. compared

four designs of equal length (20 weeks) and equal number of assessments (8 after baseline): open label (OL), open label followed by blinded discontinuation (OL+BDC), a traditional crossover (CO), and the Hybrid design (Figure 1). The Hybrid design was chosen for the trial.

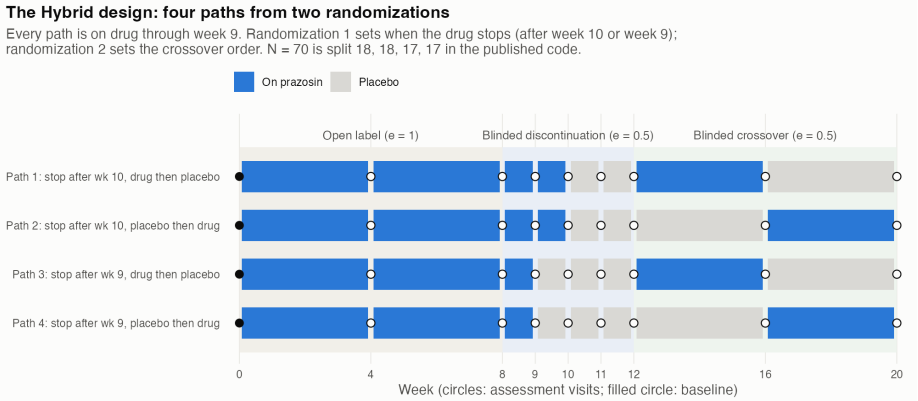


Figure 1: The Hybrid design. Each row is a path; blue intervals are on drug.

Why each phase is there.

- **Open label, weeks 0 to 8.** Every participant starts on active drug, knowing it. Hendrickson et al. included this phase so that acutely symptomatic patients, who would decline possible placebo assignment, could enroll; they report that recruitment then outpaced the budget [1]. The phase also titrates the dose; the paper takes prazosin titration to need 2.5 weeks each time the drug is started. Statistically, the open-label phase contributes on-drug visits at full expectancy ($e_t = 1$). Drug and expectancy response are both present at full strength there and cannot be separated within the phase.
- **Blinded discontinuation, weeks 8 to 12.** Participants are told they may be switched to placebo during the block. Every path is on drug at week 9. Randomization 1 then decides whether the drug stops after week 9 or after week 10, so at week 10 half the participants are on drug and half off, with expectancy matched. In the paper's schedule only the participant is blinded during this block, except in its second week, when the switch is randomized and staff are blinded as well [1]. Expectancy is 0.5 throughout the block.
- **Crossover, weeks 12 to 20.** Two 4-week blinded periods, one on drug and one on placebo, in an order set by randomization 2. Each period has one assessment at its end (weeks 16 and 20). Expectancy is 0.5.

The four paths are the two discontinuation times crossed with the two crossover orders. The published code does not randomize patients to paths:

it allocates them in fixed numbers (18, 18, 17, 17 at $N = 70$) and generates each path's data separately. The path definitions used in this project's scripts match the published `buildtrialdesign()` output for every path and every field (verified, re-fetched from commit `58b32a9`). Visit by visit:

Table 5: Drug state, t_{od} and t_{sd} by visit for the four Hybrid paths

Path	Drug state	t_{od} (weeks)	t_{sd} (weeks)
1	1 1 1 1 0 0 1 0	4 8 9 10 0 0 4 0	0 0 0 0 1 2 0 4
2	1 1 1 1 0 0 0 1	4 8 9 10 0 0 0 4	0 0 0 0 1 2 6 0
3	1 1 1 0 0 0 1 0	4 8 9 0 0 0 4 0	0 0 0 1 2 3 0 4
4	1 1 1 0 0 0 0 1	4 8 9 0 0 0 0 4	0 0 0 1 2 3 7 0

Two features of this table matter later. First, the paths differ in drug state only at weeks 10, 16 and 20. At weeks 4, 8, 9, 11 and 12 every path is in the same state. Second, on re-exposure in the crossover, time on drug restarts from zero rather than continuing from the earlier run, so it is 4 weeks at the re-exposure visit (week 16 or 20). The simulated drug response therefore rebuilds from zero.

3. The three response components

Hendrickson et al. model each participant's symptom score as baseline minus the sum of three responses [1]:

$$Y_{it} = \mathrm{BL}_i - \left[\, TV_{it} + PB_{it} + BR_{it} \,\right].$$

The paper calls the components BR, ER and TR; this project calls them BR, PB and TV. Each component has its own mean trajectory and its own clock.

Table 6: Clock, mean trajectory and SD of each response component

Component	Clock	Mean at visit t	SD
BR	time on drug t_{od}	$G(t_{od}; 10.99, 5, 0.42)$, plus carryover	8
PB	time under expectancy t_{pb}	$e_t\, G(t_{pb}; 6.51, 5, 0.35)$	$10\, e_t$
TV	calendar time w_t	$G(w_t; 6.51, 5, 0.35)$	10

- BR is the pharmacological response to prazosin. It runs only while the participant is on drug and decays after discontinuation.
- PB is the response to believing one may be on active drug. It is scaled by the design expectancy.

- TV collects everything that changes with time in the trial regardless of treatment: regression to the mean, the effect of regular contact and assessment, and natural change in the illness.

The parameters are the published `extracted_rp` set, printed directly from the `58b32a9` data. Baseline has mean 83.07 and SD 18.48 CAPS points, and the biomarker mean 124.33 and SD 15.36 mmHg.

How the parameters were obtained. The BR trajectory was fitted to the difference between the prazosin and placebo arms of the active-duty trial [2]. The placebo arm's trajectory was taken to be $TV + PB$. Because no data separate those two, the tabula rasa assumption gives them equal parameters [1]. The split between TV and PB is therefore an assumption, not an estimate (`docs/25`, Section 4; `docs/31`, Section 14).

Expectancy acts twice. The design expectancy e_t scales both the mean and the standard deviation of PB. Blinding at week 9 halves the expectancy response, and it also halves that response's variability. This is the only route by which blinding enters the data.

What differs between paths. In the Hybrid design every path has the same visit weeks, and expectancy stays positive throughout, so $t_{pb} = w_t$ in every path. Hence TV and PB have identical means in all four paths. Only BR differs, through the drug-state pattern (Figure 3; numbers in the table below, no carryover).

Table 7: Expected component means by week and Hybrid path without carryover

Week	4	8	9	10	11	12	16	20
TV, all	1.86	4.79	5.24	5.59	5.85	6.03	6.39	6.48
PB, all	1.86	4.79	2.62	2.79	2.92	3.02	3.19	3.24
BR, path 1	4.28	9.22	9.79	10.19	0	0	4.28	0
BR, path 2	4.28	9.22	9.79	10.19	0	0	0	4.28
BR, path 3	4.28	9.22	9.79	0	0	0	4.28	0
BR, path 4	4.28	9.22	9.79	0	0	0	0	4.28

Figure 3 reproduces panel D of Hendrickson's Figure 3 from the published parameters. Three things stand out:

- **The total peaks at the end of open label** (18.8 points at week 8). Drug and full expectancy are both at work there. It drops at week 9 although everyone is still on drug, because expectancy halves.
- **Discontinuation produces the largest single change.** The drug response falls from 10.2 to 0 for paths 1 and 2 after week 10, and for paths 3 and 4 after week 9.

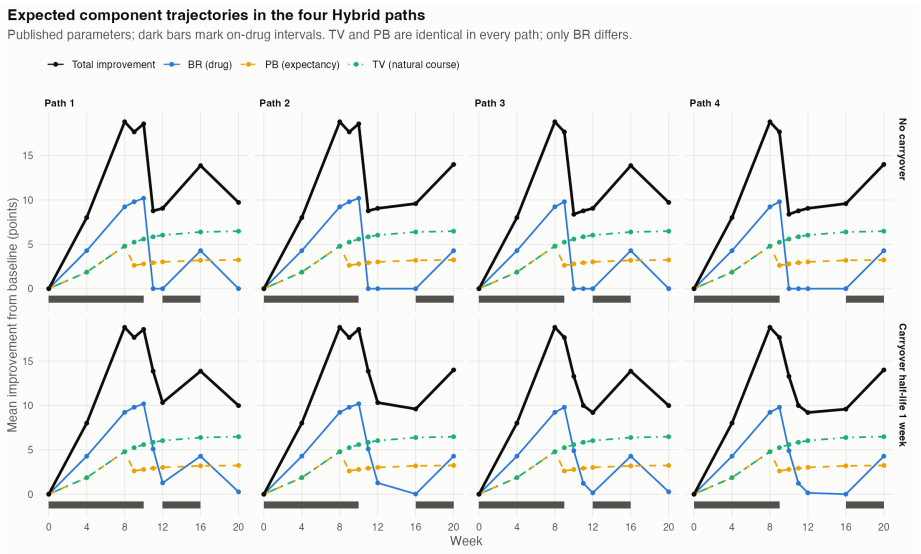


Figure 2: Expected component trajectories in each Hybrid path, without carryover (top) and with a one-week carryover half-life (bottom).

- **The crossover's on-drug period recovers only 4.3 points** of drug response, against 9 to 10 at the end of open label. Time on drug restarts at zero (Section 4).

Carryover. With $t_{1/2} > 0$, the off-drug mean of BR is computed recursively: the previous visit's mean times $2^{-t_{sd}/t_{1/2}}$, with t_{sd} cumulative. At $t_{1/2} = 1$ week, path 1 has 5.09 at week 11, half of 10.19. At week 12 it has 1.27, where decay anchored at discontinuation would give 2.55. The recursion therefore decays faster than the stated half-life (`docs/38`, Section 3). Carryover affects only BR; TV and PB are unchanged. Prazosin itself leaves plasma with a half-life of about 2.3 hours [4], so the carryover that matters on a weekly visit schedule is pharmacodynamic persistence of the response, not drug remaining in the body (`docs/38`, Section 2).

4. The modified Gompertz curve

Each component's mean is a modified form of the Gompertz curve [3],

$$G(t; m, d, r) \;=\; m\,\frac{\exp\!\left(-d\,e^{-rt}\right) - e^{-d}}{1 - e^{-d}},$$

which is the code's `y = maxr * exp(-disp * exp(-rate * t))`, shifted to pass through zero at $t = 0$ and rescaled to approach m. Each parameter has a

geometric meaning (Figure 2):

- **Ceiling** m (`max`) scales the whole curve. It is the improvement reached after a long time on drug (or in the trial, for TV).
- **Displacement** d (`disp`) sets how long the curve waits before it rises. Larger d gives a longer lag and a steeper rise once it starts.
- **Rate** r (`rate`, per week) compresses or stretches time. Doubling r halves every landmark time.

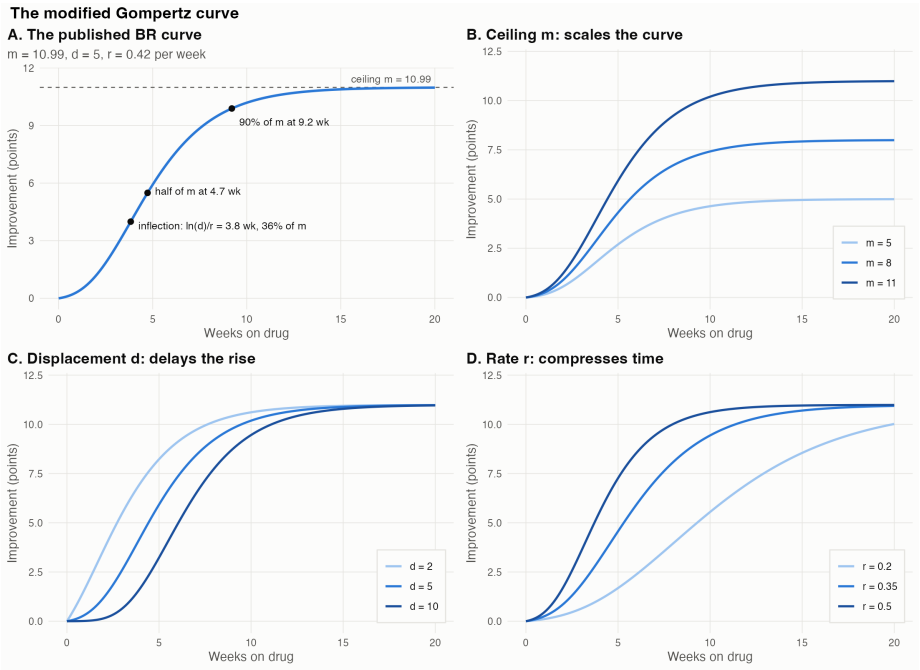


Figure 3: The modified Gompertz curve: the published BR curve with its landmarks (A) and the effect of each parameter (B to D).

Landmarks. These are computed from the formula (verified, `gompertz-landmarks.csv`). The inflection is at $t^* = \ln(d)/r$, where the curve has reached $(e^{-1} - e^{-d})/(1 - e^{-d})$ of its ceiling, 36% when $d = 5$.

Table 8: Landmarks of the modified Gompertz curves for BR and for TV and PB

Curve	Inflection	Half of m	90% of m	Share of m, weeks 1 / 4 / 8
BR	3.8 wk	4.7 wk	9.2 wk	0.03 / 0.39 / 0.84
TV, PB	4.6 wk	5.7 wk	11.1 wk	0.02 / 0.29 / 0.74

Curve	Inflection	Half of m	90% of m	Share of m, weeks 1 / 4 / 8

(BR: $m = 10.99$, $d = 5$, $r = 0.42$; TV and PB: $m = 6.51$, $d = 5$, $r = 0.35$.)

What the shape implies for the design.

- **Slow start.** After one week on drug the response is 3% of its ceiling. A blinded block that returns a participant to drug after a week off therefore observes almost no drug response at the return visit. The published coupling and mean-moderation rules treat that visit as fully moderated anyway; under moderation proportional to the response, as `docs/38` proposes, it carries almost no interaction information.
- **Restart from zero.** Because time on drug resets at every re-exposure, the crossover's 4-week on-drug period reaches only 39% of the ceiling. This is a modeling choice in the published code, not pharmacology. `docs/38` argues that the response should rebuild from its residual level.
- **Monotone, with a population ceiling.** The curve cannot represent a response that fades while drug continues. Every participant has the same ceiling; individual differences enter only through the multivariate normal noise around the mean (`docs/25`, Section 6).

How much the choice of curve matters. Manuscript 07 replaced the Gompertz with a logistic, a hyperbolic tangent and a piecewise-linear curve, matched at the ceiling and at the week-5 response, in the OL+BDC design at $N = 70$ (`analysis/report/07-gompertz-evaluation/`). Under covariance moderation the curve family has no effect at all: the interaction sits in the correlation matrix, not in the mean. With matched random-number seeds the four families gave identical rejection decisions. Under mean moderation with the shift scaled to the response trajectory, power ranged from 0.743 (Gompertz, the lowest) to 0.782 (logistic), about two Monte Carlo standard errors. The Gompertz is a defensible and mildly conservative default; it is not the main source of uncertainty in the simulation.

5. The interaction on the trajectory template

Hendrickson's Figure 3 shows the population mean of each component. The interaction is invisible there, because it concerns how responses differ between participants. Conditioning the same template on the biomarker makes it visible.

Covariance moderation. In the published construct, the biomarker is correlated with BR at visit t with correlation $c_{bm} g_t$, and with nothing else. Under the multivariate normal model the expected drug response of a participant with standardized biomarker b is

$$E(BR_t \mid b) = E(BR_t) + c_{bm}\, g_t\, \sigma_{BR}\, b,$$

and TV, PB and baseline do not depend on b. **Mean moderation** gives the same form directly, with β_{bm} in place of c_{bm} and the on-drug indicator in place of g_t. Participants one SD above and one SD below the biomarker mean therefore have expected trajectories that differ by $2c_{bm}g_t\sigma_{BR}$ at visit t: 4 points at full coupling and $c_{bm} = 0.25$. Figure 4 draws the two trajectories on the template.

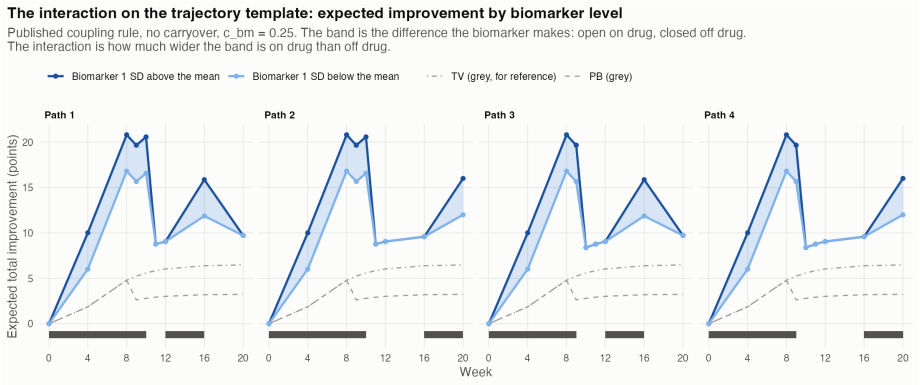


Figure 4: The template conditioned on the biomarker: expected total improvement at one SD above and one SD below the biomarker mean, published coupling, no carryover.

The band opens wherever the drug response is coupled to the biomarker and closes wherever it is not. Under the published rule without carryover, that means open at every on-drug visit and closed at every off-drug visit. The biomarker predicts improvement on drug and nothing off drug, which is a biomarker-by-drug interaction.

The estimand is a difference in band width. For one path, the E9 statistic contrasts each participant's on-drug and off-drug means. Its slope on the biomarker has expectation (`docs/37`, Section 3)

$$\beta_{bm:D} \;=\; -\,\frac{\overline{\text{gap}}_{\text{on}} - \overline{\text{gap}}_{\text{off}}}{2\,\sigma_{bm}} \;=\; -\,c_{bm}\,\frac{\sigma_{BR}}{\sigma_{bm}}\,(\bar{g}_{\text{on}} - \bar{g}_{\text{off}}),$$

where $\overline{\text{gap}}$ is the band width averaged over the on-drug or the off-drug visits. Figure 5 plots the band width visit by visit for path 1 under four coupling rules.

Per-path slopes, computed from the band widths (verified, `interaction-gap-by-path.csv`):

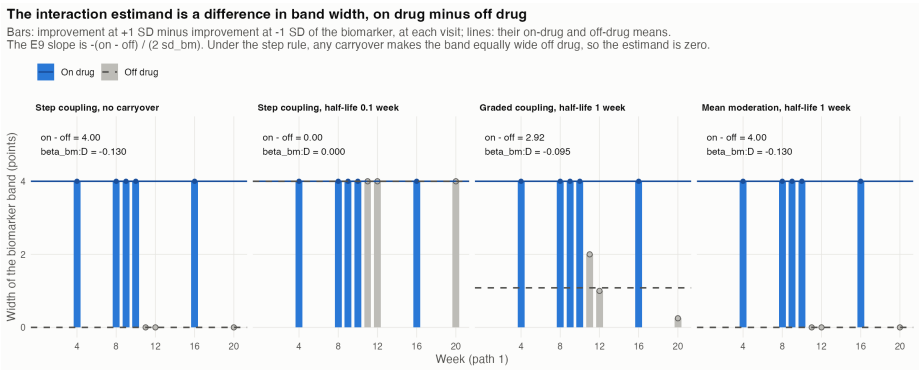


Figure 5: The band width at each visit of path 1 under four coupling rules, with its on-drug and off-drug means.

Table 9: Per-path interaction slopes by coupling rule and carryover half-life

Rule, $t_{1/2}$ (weeks)	Path 1	Path 2	Path 3	Path 4
Step, 0	−0.130	−0.130	−0.130	−0.130
Step, 0.1	0	0	0	0
Graded, 1	−0.095	−0.097	−0.100	−0.101
Mean moderation, 1	−0.130	−0.130	−0.130	−0.130

This is problem 2 in one picture. The published rule couples the biomarker wherever the mean drug response is nonzero, and with any carryover it is never exactly zero after first exposure. At a half-life of 0.1 weeks the residual drug effect one week after stopping is 0.1% of its on-drug value, yet the band is fully open at every off-drug visit (second panel). The on-minus-off difference, and with it the estimand, is then exactly zero. The power collapse that Hendrickson et al. report, from 0.74 to 0.12 at $N = 70$ and $c_{bm} = 0.3$ (`docs/36`, Section 5.1), is the analysis correctly finding nothing to detect. Under graded coupling the off-drug band narrows as the drug effect fades, and the estimand declines smoothly. Under the published mean moderation the band closes the moment the drug stops, so the estimand does not depend on the half-life.

The paths differ under graded coupling because path 4's off-drug visits lie furthest from a discontinuation (up to 7 weeks), so its off-drug band is narrowest. This is why the path whose drug stops earlier and whose crossover is placebo first carries the most interaction information under carryover.

6. The correlation structure

The published construct draws each participant's 26 variables jointly from a multivariate normal distribution: biomarker, baseline, and TV, PB and BR at

8 visits. The correlation matrix (Figure 6) has four parts:

- **Within each component,** $\rho = 0.8$ **between any two visits**, whatever the time between them. This is compound symmetry.
- **Across components,** $c_1 = 0.2$ **at the same visit and** $c_\times = 0.1$ **at different visits.**
- **Biomarker and** BR**:** c_{bm} **at the coupled visits, 0 elsewhere.** The biomarker is uncorrelated with baseline, TV and PB. There is no parameter in the software for a biomarker correlated with natural course (`docs/31`, Section 18.1).
- **Baseline is uncorrelated with every response component.**

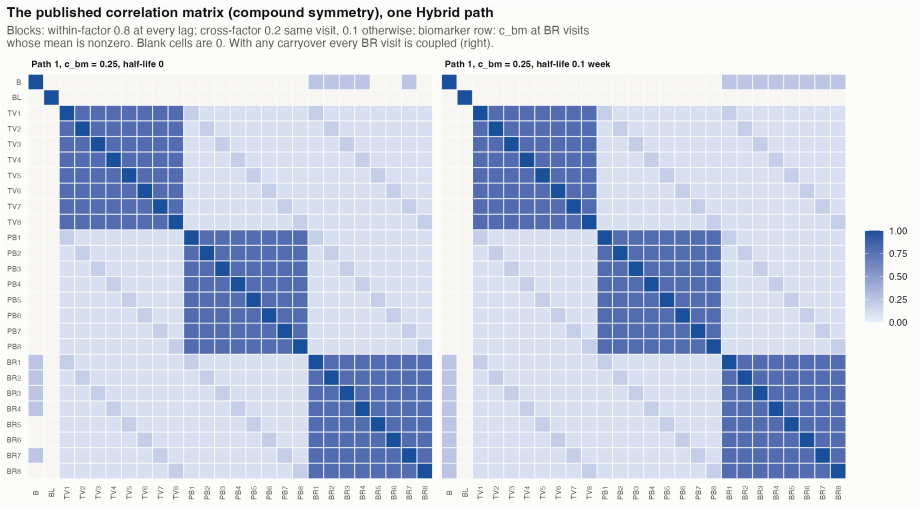


Figure 6: The published correlation matrix for path 1 at $c_{bm} = 0.25$, without carryover (left) and with a 0.1-week half-life (right).

Why this matters.

- **Problem 1.** Under compound symmetry the biomarker row can carry at most $c_{bm} = 0.256$ when the coupling has an on/off contrast (`docs/36`, Section 4.2). The published effect sizes 0.3 and 0.6 exceed that ceiling in every Hybrid path without carryover. The code then forces the matrix to be positive definite, which at $c_{bm} = 0.6$ lowers the simulated correlation to 0.54 to 0.57. With any carryover the coupling is constant (Figure 6, right), the matrix is valid, and the interaction is gone. The two problems are the two sides of the step rule (`docs/36`, Section 5.4).
- **The analysis has to match the covariance of the outcome.** Even under the published compound-symmetry construct, the outcome is not

compound symmetric. The covariance of Y computed from the construct (path 1, no interaction) has three blocks:

Table 10: Covariance blocks of the outcome Y under the compound-symmetry construct, path 1

Visits	Variance	Covariance within	Covariance with baseline
Baseline	342		
Open label (weeks 4, 8)	710	605	342
Blinded (weeks 9 to 20)	599	527	342

The covariance between open-label and blinded visits is 556. Baseline carries no noise of its own (its value is BL_i exactly), so its covariance with every later visit equals its variance. The published random intercept forces all of this into one between-participant variance and one residual variance. If the data are AR(1) instead, the published analysis rejects 14% to 23% of null crossover trials, and a `corCAR1` residual structure restores the size (`docs/36`, Section 6.8).

7. What the design identifies

The design's value lies in which comparisons it makes possible. Since TV and PB are the same in every path at a given visit, any comparison between paths at the same visit isolates BR: a randomized comparison, untouched by natural course and expectancy. In the Hybrid design such comparisons exist at weeks 10, 16 and 20 (and, under carryover, at weeks 11 and 12, where the time since stopping differs). Comparisons within a path, between on-drug and off-drug visits, are far more numerous but rest on modeling time and expectancy.

- **Between paths at the same visit** (weeks 10, 16, 20): identifies BR, relying on the randomization only.
- **Within a path, open label against blinded, on drug** (weeks 8 and 9): identifies the expectancy drop, $0.5\,PB$, relying on the expectancy model.
- **Within a path, on drug against off drug** (all visits): identifies BR plus any difference in time and expectancy between the visits compared, relying on the time and expectancy terms.
- **Any of these by the biomarker**: identifies $\beta_{bm:D}$, relying on the biomarker being unrelated to TV and PB.

The interaction is protected by an assumption, the drug effect by the randomization. The interaction is estimated from how the within-person on-minus-off contrast varies with the biomarker. Because the biomarker is unrelated to TV and PB in the simulation, any misfit of the time and expectancy terms

adds noise to this contrast but no bias. The drug main effect has no such protection. It is biased by any time or expectancy trend that the model fails to capture, unless it is estimated from the between-path comparisons alone (Section 9).

Two consequences.

- **The mixture of paths does not affect interaction power in Hybrid.** Putting all 70 participants on one path changes `lmer` power by at most 0.04 from the published allocation (`docs/37`, Section 6; verified, 1000 replicates per cell). The randomization is not what powers the interaction test here. What it buys is protection against a biomarker associated with natural course or expectancy, which the simulation rules out by construction, and identification of the drug main effect.
- **The design's limit on expectancy.** With one open-label and one blinded level of expectancy, the data identify only the product of the expectancy weight and the PB ceiling. The value 0.5 is an assumption (`docs/31`, Section 7.3).

8. Building the analysis model

The simulation's mean structure, for a participant in path p with standardized biomarker b, is

$$E(Y_{it} \mid b) = \mu_{\text{BL}} - TV(w_t) - e_t\, PB(t_{pb}) - BR_p(t) - [\text{moderation}]_t\, b.$$

An analysis model replaces each part with something estimable. The published model, under the publication settings (`useDE = FALSE`, `t_random_slope = FALSE`; `58b32a9` vignette 1, line 350), is

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_t w_t + \beta_{bm:D}\, B_i D_{it} + u_i + \varepsilon_{it},$$

in code `Sx ~ bm + Db + t + bm*Db + (1|ptID)`, fitted with `lmer`. The baseline enters as a row at $t = 0$ with $D = 0$. Term by term:

- $\beta_0 + u_i$**: baseline level and stable individual differences.** The random intercept matches compound symmetry. It ignores the change in variance at blinding.
- $\beta_B B_i$**: a prognostic biomarker effect,** zero in the simulation. The biomarker is uncentered (mean 124), so β_D is the drug effect at a biomarker of zero. Centering fixes the interpretation and leaves $\beta_{bm:D}$ unchanged.
- $\beta_D D_{it}$**: the drug response** $BR_p(t)$**.** One number for a response that is 4.3 at week 4, 10.2 at week 10 and 4.3 again after re-exposure.

- $\beta_t w_t$**: the non-drug trends** $TV(w_t)+e_t PB(t_{pb})$**.** A straight line through two sigmoids and a step. It is significant in every fit (Section 10).
- $\beta_{bm:D} B_i D_{it}$**: the moderation,** the estimand.
- $\beta_e e_t$ **(`De`): the expectancy step.** Present in the code, switched off for the publication.

The expectancy term. Hendrickson et al. state that including expectancy 'was found to increase the frequency of collinearity leading to poor model fits while changing results minimally' [1]. In the Hybrid long data (baseline rows included, weighted by allocation), `De` has correlation 0.59 with `Db`, and $R^2 = 0.35$ on `Db` and `t` together (variance inflation 1.5; verified). Every open-label visit is on drug, and every visit with `De = 0` is baseline. The collinearity is real but moderate. With $N = 70$ the term is estimable; its cost is precision, and it changes the drug effect (Section 9).

Time as a factor. Replacing $\beta_t w_t$ by one free mean per visit, τ_t (code `factor(t)`), absorbs everything common to all participants at a visit. That includes TV, PB, the expectancy step, and the part of BR that every path shares. At weeks 4, 8 and 9 every path is on drug with the same time on drug, so the Gompertz rise of BR during open label goes into τ_t, not into β_D. The drug effect is then identified only by the between-path comparisons of Section 7: a randomized estimate, but a less precise one. With a single path, D and τ_t are completely aliased. **A spline in time** (`ns(t, 3)`) is the intermediate choice. It captures the sigmoids but not the step at week 9, and it leaves the within-path on/off variation available for β_D.

Other choices, from earlier work.

- **Exposure coding.** E1 is the binary D_{it} (published). E2 is the exposure-decayed $D_{bc,it}$, which assumes a decay form and a half-life. E3 adds a lagged just-off-drug term (NOTATION.md, Part 3).
- **Residual correlation.** `corCAR1` is required when the data are serially correlated (`docs/36`, Section 6.8).
- **Path term.** A fixed path effect changes `lmer` interaction power by at most 0.001. The random intercept already covers it (`docs/37`, Section 6).
- **Open-label branch.** When no participant varies in drug state (the OL design), the code fits `Sx ~ bm + t + bm*t + (1|ptID)` and tests the biomarker-by-time slope. That slope is fully confounded with natural course and expectancy.
- **What 'unstructured' refers to.** The paper says models were 'run with maximal random effects structure justified by the design' and that 'an unstructured variance/covariance matrix was assumed' [1]. The code fits every model with `lme4::lmer`, with random effects `(1|ptID)` or `(1+t|ptID)` and independent, constant-variance residuals. It has no correlation or variance-weights argument (inspected, `lme_analysis.R` at `58b32a9`, lines 106 to 137). Read with the reference to maximal random effects, 'unstructured' most plausibly describes the covariance of the random effects,

which `lmer` estimates freely for `(1+t|ptID)` (inferred). The published runs used `(1|ptID)`, so the fitted model implies compound symmetry for the repeated measurements. It is not an unstructured outcome covariance in the MMRM sense.

Summary measures and repeated-measures ANOVA. The E9 statistic of `docs/37` is the two-stage summary-measures counterpart of this model: an on-minus-off contrast per participant, then a regression on the biomarker. It equals the interaction test of a repeated-measures ANCOVA with drug state as a two-level within-subject factor. It has no time term, which is why it loses power in the crossover, where the time trend enters the two sequences' contrasts with opposite sign (`docs/37`, Section 5.7). `docs/26` derives the strict repeated-measures ANOVA version with a dichotomized biomarker.

9. Does the form of the time adjustment matter?

Three forms of time adjustment were compared on the same simulated trials: linear `t` (published), `ns(t, df = 3)`, and `factor(t)`. The setting was the published construct (CS), $N = 70$, strength 0.25, the published allocation, and three coupling rules at $t_{1/2} \in \{0, 1, 2\}$ weeks, in Hybrid and CO. There were 500 replicates per cell and 1000 per null cell, and all 39,000 fits converged (script `07-time-adjustment.R`; verified).

Population means first. A weighted least-squares fit of each form to the true population means shows what each form can represent (Figure 7). Linear time leaves a root-mean-square misfit of 3.2 points, adding `De` leaves 3.1, the spline 1.9, and visit as a factor 0.8. What visit as a factor still misses is the drug response that one `Db` coefficient cannot represent (10.2 at week 10 against 4.3 after re-exposure).

The drug effect depends on the time adjustment. The same population-mean calculation gives the drug coefficient each form converges to. Simulation agrees with it (Hybrid, graded coupling; Monte Carlo mean and SD over 500 replicates):

Table 11: Limiting and simulated drug coefficient β_D by time adjustment

Time adjustment	Limit, $t_{1/2} = 0$	Simulated	Limit, $t_{1/2} = 1$	Simulated
Linear `t`	−8.21	−8.16 (0.86)	−6.78	−6.70 (0.84)
Linear `t` + `De`	−7.28	not run	−5.34	not run
Spline	−6.22	−6.02 (0.99)	−3.97	−3.78 (0.94)
Visit as factor	−6.25	−5.99 (1.27)	−4.52	−4.28 (1.19)

'Limit' is the population-mean value of β_D; 'simulated' is the Monte Carlo mean with its SD in parentheses.

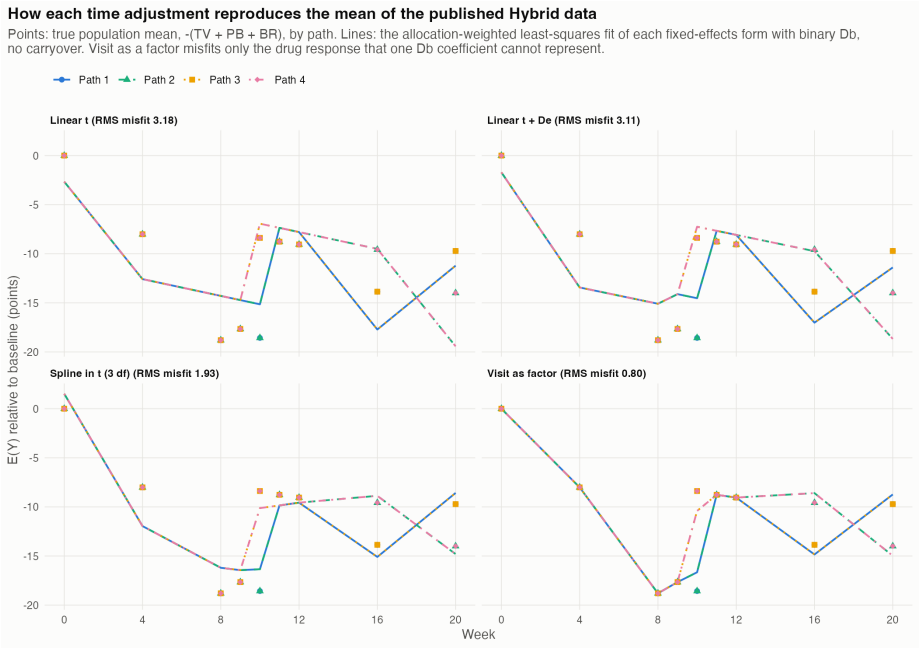


Figure 7: True population means by path (points) and the best fit of each time adjustment (lines), no carryover.

The linear-`t` drug effect is about 2 points larger than the visit-factor effect, with or without carryover (2.0 and 2.3 points in the population-mean limit, 2.2 and 2.4 in simulation). The open-label visits carry both drug and full expectancy, the expectancy then drops by half at week 9, and a straight line cannot follow the drop. The excess open-label improvement is therefore credited to the drug. Adding `De` removes half to two thirds of the difference (0.93 of 1.96 points without carryover, 1.44 of 2.26 at a one-week half-life). In the crossover, where expectancy is constant, the three forms agree within about 0.5 points (CO, graded, $t_{1/2} = 0$: -6.82, -6.29, -6.74, a largest difference of 0.53; verified, `summary-reps500.csv`).

None of these numbers is 'the' drug effect. Each form targets a different weighted average of a drug response that varies from 4.3 to 10.2 across visits. Visit as a factor targets the between-path contrasts at weeks 10, 16 and 20 alone, and pays for that in precision (SD 1.27 against 0.86).

The interaction does not depend on the time adjustment. In every cell the three mean estimates of $\beta_{bm:D}$ agree to within 0.0007. In Hybrid their empirical SDs also agree (graded, $t_{1/2} = 0$: 0.0562, 0.0559, 0.0558). This is what Section 7 predicts: misfit of the time terms is unrelated to the biomarker.

Power and size. Rejection rates of the interaction test:

Table 12: Interaction test rejection rates by design, cell and time adjustment

Design	Cell	Linear `t`	Spline	Visit as factor
Hybrid	null, $t_{1/2} = 0$	0.049	0.055	0.059
Hybrid	null, $t_{1/2} = 1$	0.058	0.073	0.080
Hybrid	graded, $t_{1/2} = 0$	0.636	0.678	0.690
Hybrid	graded, $t_{1/2} = 1$	0.444	0.484	0.508
Hybrid	mean moderation, $t_{1/2} = 2$	0.626	0.682	0.696
CO	null, $t_{1/2} = 0$	0.052	0.040	0.044
CO	null, $t_{1/2} = 1$	0.042	0.039	0.039
CO	graded, $t_{1/2} = 0$	0.734	0.750	0.772

Monte Carlo standard errors are about 0.007 for the null rates and 0.021 for power.

- **In Hybrid, the flexible forms gain power partly by losing size.** Their estimates are no less variable than linear time's, yet they reject more often, so their model-based standard errors must be smaller. Under the null with one-week carryover their Type I error is 0.073 and 0.080, about 3 and 3.5 Monte Carlo standard errors above the nominal 0.05 (standard errors at the observed rates, 0.0082 and 0.0086; verified, `summary-reps1000-null.csv`), against 0.058 (about 1) for linear time. Without carryover all three lie within 1.2 standard errors of 0.05. The

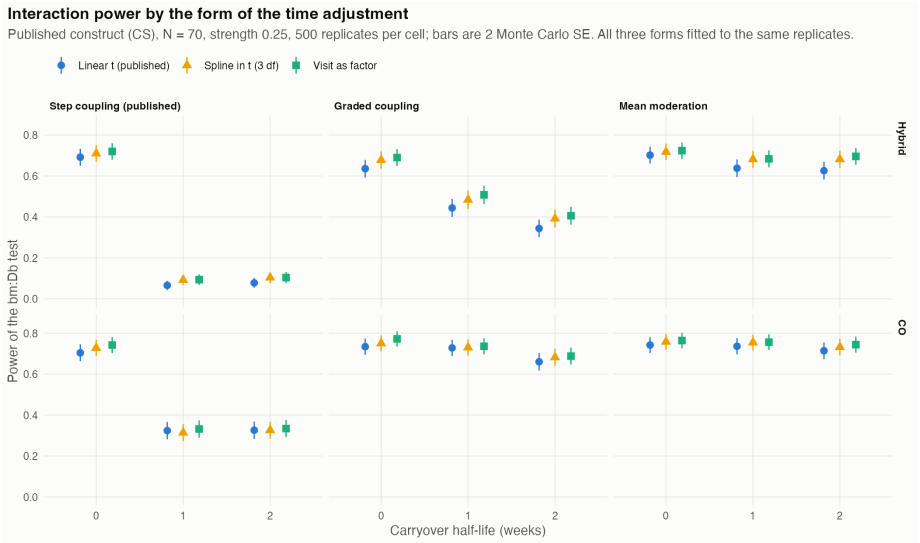


Figure 8: Interaction power by the form of the time adjustment, with 2-SE bars.

power gains (0.02 to 0.07) are therefore not clean improvements. A size-adjusted comparison was not made.

- **Why the flexible forms understate the standard error is not established.** A plausible mechanism (inferred, not tested): the data's variance changes at blinding, because the PB standard deviation halves, and the drug response the single `Db` cannot represent varies by path and visit. The random-intercept model assumes neither, and the residual misfit left by linear time happens to inflate its error variance enough to offset the understatement.
- **In the crossover all three forms hold their size**, and the flexible forms gain 0.01 to 0.04, partly from a real reduction in the estimate's variability (SD 0.0487 against 0.0462).

10. Recommendations

For the interaction test in a Hybrid trial:

1. **Keep a time term.** It is highly significant in every fit (slope −0.43 to −0.44 points per week in Hybrid, median p about 10^{-8}; verified). Leaving it out confounds the drug state with natural course.
2. **Linear time is adequate for the interaction and holds its size** in this construct. Do not switch to a spline or visit-as-factor to gain power without also fitting a residual structure that reflects the variance change at blinding (for example `weights = varIdent(form = ~ 1 | De)` in `nlme`)

and the serial correlation (`corCAR1`), and without checking the size. That combination has not been tested.
3. **Center the biomarker**, so that the drug coefficient is the effect at the average biomarker.

For the drug main effect:

4. **Model expectancy or use visit effects.** Linear time without `De` overstates the drug effect by about 2 points in Hybrid, because the open-label expectancy is credited to the drug. Visit-as-factor gives a randomized estimate from the between-path comparisons, at about 1.5 times the standard deviation.

For the simulation:

5. **Use a coupling rule that fades with the drug effect** (graded, or moderation proportional to the response). The published step rule is the cause of both known problems (Sections 5 and 6).
6. **Treat the Gompertz parameters, the tabula rasa split of TV and PB, the expectancy weight 0.5 and the restart from zero as assumptions to vary**, not as properties of prazosin.

11. Corrections to earlier documents

The synthesis turned up errors in earlier project documents. Each statement below was checked against the published code, data or paper.

1. **`docs/25`, Section 4 table.** It gives TV the parameters $m = 10.99$, $d = 5$, $r = 0.42$. Those are BR's. TV has the PB parameters, $m = 6.51$, $d = 5$, $r = 0.35$, SD 10 (`extracted_rp` at `58b32a9`).
2. **`docs/31`, Sections 5 and 7.1, and `docs/24`.** They describe the crossover as open label with expectancy 1. It is blinded, with expectancy 0.5 (Hendrickson Figure 2D; `e` in the vignette).
3. **`docs/31`, Sections 3 and 10.** They describe a sixteen-week trial with the switch to placebo at the start of week 9. The trial lasts 20 weeks; everyone is on drug at week 9, and the drug stops after week 9 or after week 10.
4. **`docs/31`, Section 2.** It says $r = 0.42$ gives 'half-maximum at about eighteen days'. Half of the ceiling is reached at 4.7 weeks, 33 days.
5. **`docs/31`, Section 6.** It gives the Hendrickson analysis as `Sx ~ bm + t + Dbc + bm:Dbc` with `corCAR1`. That is the later vendored implementation. The published `58b32a9` analysis is `lmer` with binary `Db`, a random intercept and no residual correlation.

6. **`docs/31`, references.** It cites Hendrickson et al. (2020) as 'Optimizing aggregated N-of-1 trial designs for the detection of biomarker-treatment interactions'. The published title is 'Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation: Statistical Methods and Theoretical Findings' [1].

`docs/31` and `docs/24` should be corrected at the source. That has not been done here.

12. Limitations

- **One construct.** Section 9 uses the published compound-symmetry construct. Under AR(1) data the conclusions about size would differ; the published analysis is then anticonservative in CO (`docs/36`).
- **No size-adjusted power.** The time-adjustment comparison reports raw rejection rates, and the null cells cover two half-lives only.
- **The drug-effect comparison has no single truth.** Each time adjustment targets a different weighted average of a time-varying drug response; Section 9 compares them with each other and with their population-mean limits, not with a true value.
- **The mechanism of the anticonservatism is a hypothesis.** It was not tested by refitting with a heteroscedastic residual structure.
- **Figures 4 and 5 show expectations.** They are exact for the simulation model; individual trajectories scatter widely around them (SDs of 8 to 10 points per component).

13. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/07-time-adjustment.R \
  --reps 500 --cores 8
Rscript analysis/scripts/quick-sim/carryover-closed-form/07-time-adjustment.R \
  --reps 1000 --cores 8 --null
Rscript analysis/scripts/quick-sim/hybrid-design-primer/01-figures.R
```

Run from the repository root. The first two write `analysis/data/quick-sim/carryover-closed-form/time-` and `summary-reps1000-null.csv` (about 15 and 12 minutes on 8 cores). The third writes Figures 1 to 8 to `docs/figures/40-fig*.png` and the supporting tables (Gompertz landmarks, component means, population-mean drug coefficients, per-path interaction slopes) to `analysis/data/quick-sim/hybrid-design-primer/`. All three read the published construct from `analysis/scripts/quick-sim/hendrickson-problems/04-power` whose design paths and parameters match the `58b32a9` code.

14. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013. Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.
2. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry* 2013; 170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133.
3. Gompertz B. On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies. *Philosophical Transactions of the Royal Society of London* 1825; 115:513-583. doi:10.1098/rstl.1825.0026.
4. Hobbs DC, Twomey TM, Palmer RF. Pharmacokinetics of prazosin in man. *Journal of Clinical Pharmacology* 1978; 18(8-9):402-406. doi:10.1002/j.1552-4604.1978.tb02456.x.